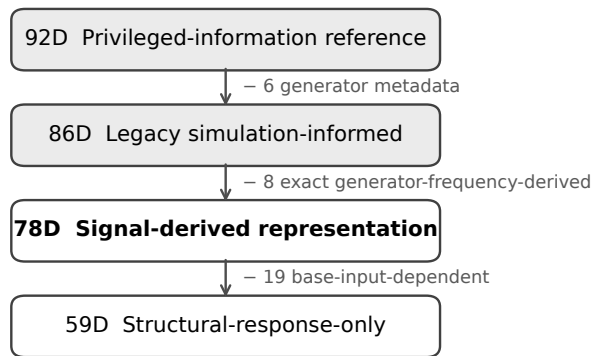


(a)



Primary manuscript representation: 78D

(c)

f_j **observable under layout S iff $D_j \subseteq S$**

D_j : structural sensors required by descriptor j

Ground-only descriptor	$D_j = \emptyset$
Story i statistics / spectrum	$D_j = \{i\}$
Story 1 relative-to-lower	$D_j = \{1\} + \text{base input}$
Story $i > 1$ relative-to-lower	$D_j = \{i - 1, i\}$
Adjacent ratio / correlation	$D_j = \{i - 1, i\}$
Spatial response fraction	$D_j = \{1, 2, 3, 4\}$

No zero masking • No descriptor redefinition

(b)

Nested provenance decomposition of the 92D space

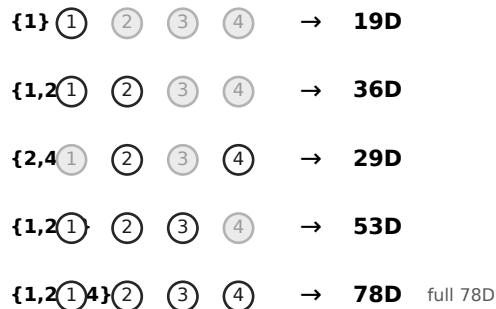
$59 + 19 + 8 + 6 = 92$ descriptors



- Structural-response-derived (59)
- Base-input-dependent (19)
- Generator-frequency-derived (8)
- Generator metadata (6)

(d)

Representative dependency-aware layouts



Base-input sensor assumed available for the primary 78D protocol